UNIVERSITY OF
CAMBRIDGE

NATURAL SCIENCES TRIPOS Part IB NST1

May 2021

Paper Biology of Disease (2) BIO/2

You have three hours to complete the paper

This is an open-book paper

*Answer **two** questions from Section A and **two** questions from Section B*

*The answer to **each** question must be written in a separate document*

Biology of Disease (2) BIO/2

SECTION A

1. Discuss how macrophages participate in the control of and recovery from infection by different pathogens.

2. Describe how allo-antigens are presented and recognised by the host immune system after solid organ transplantation.

3. Compare the replication strategies of influenza virus, herpesviruses and HIV. Comment on how anti-viral drugs inhibit the replication of these three viruses.

4. Discuss how viruses modify the infected cell and how these modifications benefit the virus.

SECTION B

5. How do bacteria use protein toxins and effectors to manipulate the eukaryotic cytoskeleton to promote their colonization, survival and dissemination in the host?

6. Compare and contrast anaemias caused by impaired generation of red blood cells or their constituents versus those caused by increased destruction of red blood cells.

7. Discuss cancer development as an evolutionary process and how this is driven by environmental factors and genetic changes.

8. Compare mechanisms of immunopathology associated with severe disease in human malaria (*Plasmodium falciparum*) and schistosomiasis (*Schistosoma mansoni*).

END OF PAPER